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GATE 2008, ECE Question Number 8

Abstract

This project implements and verifies a logic function derived from a GATE 2008 Karnaugh map (K-map) question using a Raspberry Pi Pico and basic components such as push buttons and an LED. The function is realized in hardware and validated through practical observation.

1. Question and Options

(A) QS + PRS + PQR + PRS + PQR

(B) QS + PRS + PQR + PRS + PQR

(C) PRS + PQR + PRS + PQR

(D) PRS + PQR + PRS + PQR

Correct Answer: (A)

2. Explanation and Simplification

- Group RS = 01 with PQ = 00, 01, 11 → QS
- Group PQ = 01, RS = 11, 10 → PRS
- Group PQ = 01 and 11 with RS = 01 → PQR
- Group PQ = 11, RS = 01, 11 → PRS
- Group PQ = 11 with RS = 01 → PQR

$$f = QS + PRS + PQR + PRS + PQR$$

3. Components

Component	Qty
Raspberry Pi Pico	1
Push Buttons (P, Q, R, S)	4
LED for Output f	1
Resistors (220Ω + 10kΩ)	5
Breadboard	1
Jumper Wires	12
Laptop with Thonny IDE	1

Table 1: List of components used

4. Setup and Connections

- Connect P to GP14, Q to GP15, R to GP16, and S to GP17.
- Connect the output LED to GP13 with a 220Ω resistor.
- Use 10kΩ pull-down resistors for each button input.
- Connect 3.3V and GND from the Pico board.

5. Logic Summary

- Inputs: P, Q, R, S (pressed = 1)
- Logic Expression: $f = QS + PRS + PQR + PRS + PQR$
- Output f is HIGH (LED ON) when expression evaluates to 1

6. GPIO Pin Mapping

- **P** – GP14 (Input)

- **Q** – GP15 (Input)
- **R** – GP16 (Input)
- **S** – GP17 (Input)
- **LED (f)** – GP13 (Output)

7. Truth Table and Verification

P	Q	R	S	f
0	0	0	1	1
0	1	0	1	1
0	1	1	1	1
1	1	0	1	1
1	1	1	0	1
1	0	1	1	1

Table 2: Input combinations that satisfy f = 1

8. Experiment Photo

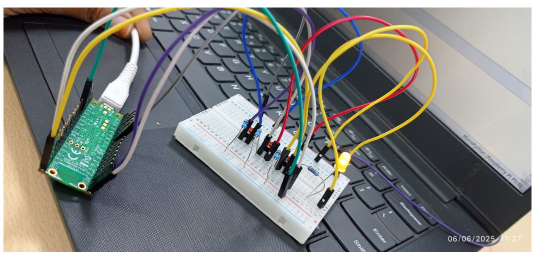

Figure: Implementation using Raspberry Pi Pico and breadboard

9. Procedure to Upload Code (Pico2W)

1. Connect Pico 2W to USB while holding BOOTSEL.
2. Drag and drop MicroPython firmware to mount as a USB drive.
3. Open Thonny IDE and select Raspberry Pi Pico as the interpreter.
4. Write and upload the logic implementation code.
5. Run and validate the LED output based on button inputs.

10. Conclusion

This experiment successfully validates the SOP logic function $f = QS + PRS + PQR + PRS + PQR$ from a Karnaugh map. The LED output matched expected truth table values, confirming proper logic implementation using Raspberry Pi Pico and GPIO inputs.

11. References

- GitHub Repository: `https://github.com/Rithrishareddy/RITHRISHA_FWC/tree/main/Hardware`
- GATE 2008 ECE Question Paper
- Raspberry Pi Pico Documentation